Abstract

Strikis, Browne and Beverland proved that no non-interleaved single-ancilla syndrome-extraction circuit for the gross code $[[144, 12, 12]]$ reaches circuit distance 12, and asked whether any interleaved circuit can. We give one. Each check's CNOT order is obtained from the order of their distance-11 circuit by reversal and in-pair swaps, operations that leave the extended code (and hence its distance 11) unchanged, while X- and Z-check CNOTs are interleaved and consecutive rounds overlap in time. Under standard circuit-level depolarizing noise (SD6) we prove $d_{\text{circ}} = 12$ in both memory bases for any number of rounds: the extended code excludes weight ≤ 10; none of the 4224 weight-11 extended-code logicals per basis lifts to an undetected circuit error; and a splitting argument reduces all round numbers to finitely many. The price is time: the period is 18 steps per round, against 8 for the earlier circuits. Within colour-symmetric schedules of this class, 18 is the minimum period for pipelined and 21 for non-wrapping schedules (in the many-round limit), and the class itself is the only one with extended-code distance 11 among 24 translation-coset colourings. Whether period 8 is possible with less symmetry is open, and the distance gain is a guarantee of the circuit rather than a demonstrated logical-error-rate gain. We also prove the conjectured $d_{\text{circ}} = 11$ of the original circuit and give exact circuit distances for 80 morphing circuits, 8 of which fall below their end-cycle distance.

A single-ancilla syndrome circuit with circuit distance 12 for the gross code

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Introduction. Bivariate-bicycle (BB) codes [1] are leading candidates for low-overhead quantum memory, and the gross code $[[144, 12, 12]]$ is the flagship of IBM's architecture [1, 6]. What a syndrome circuit delivers, however, is not the code distance d but the circuit-level distance d_{circ}: the minimum number of circuit faults that flip a logical observable without triggering any detector. For the gross code no circuit had been shown to have $d_{\text{circ}} = d$. The IBM circuit (8 time steps per round) has $d_{\text{circ}} \leq 10$ [1, 6]. Strikis, Browne and Beverland (SBB) [2] proved that every *non-interleaved* single-ancilla circuit has $d_{\text{circ}} \leq 11$ (their Prop. 4, which carries no depth assumption), gave a depth-8 left-right circuit (their Fig. 12) conjectured to reach 11, and stated as open (Sec. IV.3) "whether any interleaved circuit can achieve circuit distance 12" [2].

We answer this question in the affirmative[C40]. Table 1 specifies a single-ancilla, CNOT-based, interleaved circuit for the gross code itself; we prove $d_{\text{circ}} = 12$ for it under SD6 noise in both memory bases and for every number of rounds[C41]. The question as posed carries no depth condition, and our circuit pays for its distance in time: its period is 18 steps per round, against 8 for the circuits of Refs. [1, 2][C44]. Within the colour-symmetric families we searched, 18 cannot be improved[C45]; whether $d_{\text{circ}} = 12$ is attainable at period 8 with less symmetry remains open.

The second half of the Letter explains why this took more than a search. We prove SBB's conjecture that Fig. 12 has $d_{\text{circ}} = 11$[C10], show that retimings which keep its CNOT orders are obstructed in every family we could decide[C14,15,16], and that among the colourings we could enumerate the extended code of Fig. 12 is the only one with $d_{\text{ext}} = 11$[C46]. Exact results for 80 morphing circuits [3], including $d_{\text{circ}} = 12$ for a different $[[144, 12, 12]]$ code[C1], show that circuit distances inferred from code properties or unrefuted upper bounds can be wrong[C5,6].

Setting. We consider memory experiments: data initialisation, R rounds of syndrome extraction, and transversal data readout, in both the Z and X basis. Except for the morphing circuits discussed later, every check has one dedicated ancilla, reset, coupled by six CNOTs and measured once per round; a round of a check may overlap in time with the previous round of other checks (pipelining). Faults are single-operation Pauli components of standard circuit-level depolarizing noise (SD6 [9]: two-qubit depolarization after each CNOT, flips after resets and before measurements, single-qubit depolarization on idle qubits), counted as distinct detector-error-model (DEM) mechanisms [8]. d_{circ} is the minimum number of such faults whose combined effect triggers no detector and flips a logical observable; we report it per basis, and in every case below the two bases agree. All lower bounds also hold for any noise model whose faults are a subset of these (e.g. no idle noise). Correlated multi-operation faults, leakage, biased noise and CZ-based compilations are outside scope. Upper bounds are realised by explicit witnesses injected with probability one into the noiseless `stim` circuit (zero detectors fired, an observable flipped).

Extended code. For single-ancilla CNOT circuits SBB append to the parity-check matrix one column per "residual" (a proper suffix of a check's CNOT order, which an ancilla fault after the k-th CNOT spreads onto the data), giving an extended code of distance d_{ext}; every single fault, propagated to the end of the circuit, has a data image equal to zero or to one extended-code column (their Table I). For the gross code with the orders of Fig. 12, $d_{\text{ext}} = 11$ in both bases[C11].

Construction. Two facts constrain the design. First, by Prop. 4 of SBB the circuit must be interleaved. Second, the CNOT orders of Fig. 12 are a dead end as far as we can decide: every one of the 4224 weight-11 extended logicals per basis lifts to an undetected circuit error of Fig. 12 itself[C13], and in every family of retimings of the Fig. 12 orders we could decide, a "ten data faults plus one hook" weight-11 error survives (below)[C14,15,16]. Changing orders freely, on the other hand, risks lowering d_{ext} below 11, after which no lift argument can certify 12.

The way out is that the extended code depends on far less than the order. A residual of length 5 equals the first qubit up to the check itself, length 4 the first two, length 3 the complement of the first three, and length 2 the last two. The extended code is therefore fixed by three unordered sets per check: the first two, the first three and the last two qubits. Swapping positions 1 and 2, swapping positions 5 and 6, and reversing the order preserve these sets, so all eight orders they generate share the extended code of Fig. 12; we verified equality of the extended codes column by column for every circuit below[C32,42]. Within this class, $d_{\text{ext}} = 11$ and the complete list of 4224 weight-11 extended logicals per basis carry over unchanged, but the times at which residuals are deposited on the data, relative to the other checks' readouts, are different, and these times decide whether an extended logical lifts.

We searched colour-symmetric schedules, choosing per colour class one of the eight orders and the CNOT times with a SAT solver, subject to legality, determinism, and constraints that block every modelled weight-11 extended logical in a bulk Pauli-frame model (SM S7); candidates were then certified exactly. Our first certified circuit had all ancillas reset and

Table 1: The period-18 gross-code circuit (`cs_P18_H24`). Checks are coloured by $(x + y)$ mod 3 of their lattice position; every check of a colour class uses the same reset step, CNOT order, CNOT steps and measurement step. Labels: $A = x^3 + y + y^2 \mapsto A_1A_2A_3$, $B = y^3 + x + x^2 \mapsto B_1B_2B_3$ (conventions in SM S2). Steps are modulo 18 and listed in CNOT order; a window that passes step 17 continues in the next period (e.g. X_1: reset at 15, first CNOT at 16, remaining CNOTs at 3–13 of the next period, measurement at 14), so consecutive rounds overlap. Right column: the map from the Fig. 12 order of the same class (rev: reversal; s_{12}, s_{56}: swap positions 1,2 or 5,6).

class	reset	CNOT order	CNOT steps	meas.	from Fig. 12
X_0	17	$A_1A_2A_3B_3B_1B_2$	0, 1, 2, 8, 9, 15	16	rev s_{56}
X_1	15	$A_1A_2A_3B_1B_3B_2$	16, 3, 4, 5, 10, 13	14	rev
X_2	17	$A_3A_1A_2B_3B_1B_2$	0, 1, 5, 6, 7, 15	16	rev s_{56}
Z_0	17	$B_2B_1B_3A_2A_1A_3$	0, 7, 8, 9, 13, 14	15	rev s_{56}
Z_1	2	$B_2B_3B_1A_3A_2A_1$	3, 4, 9, 10, 11, 17	0	rev $s_{12}s_{56}$
Z_2	16	$B_2B_1B_3A_3A_2A_1$	17, 5, 6, 12, 13, 14	15	rev s_{56}

measured together (period 21)[C31]; allowing rounds to overlap gave the period-18 circuit of Table 1 and Fig. 1, and a second, different period-18 circuit with the same certified distance[C41,43]. Table 1 is interleaved, legal and deterministic and measures the gross code itself[C41,42]. In the non-wrapping family of the first search, 0 of 233 random legal deterministic circuits certify, whereas the constraint-guided search produced the certified circuits[C39]: the distance comes from the blocking reasoning, not from the class alone.

Proof of $d_{\text{circ}} = 12$. *Weight ≤ 10.* Augment the circuit with one observable per row of the extended code's check matrix H and logical matrix L, evaluated on the final data readout. In our circuit template, the augmented observable of row H_i equals the sum of all detectors of check i (first round, consecutive-round comparisons, final readout). Hence for any undetected fault set the H parts of its images sum to zero and its logical effect is the L part, so its nonzero images form an extended-code logical of weight at most the number of faults. This uses only this template identity and SBB's Table I property, not their Prop. 3, and is therefore insensitive to interleaving and pipelining. Both ingredients were checked on the DEM, in both bases, for every R used below, without exception; the image set equals the 360 extended columns[C33,42]. With $d_{\text{ext}} = 11$, no undetected logical of weight ≤ 10 exists.

Weight 11. A weight-11 undetected logical must map to 11 distinct columns forming one of the 4224 weight-11 extended logicals of its basis. For each of them we decide exactly, with CP-SAT, whether one DEM term per column can be chosen with zero detector sum. None lifts: 0/4224 in both bases for every $R = 1, \ldots, 32$; the same program lifts all 4224 for Fig. 12 (positive control)[C42].

All R. Pipelining removes the clean separation between rounds, so we use a gap argument that needs no assumption on the schedule beyond its period. After the last fault of a fault set, the data frame settles within $D = 18$ layers (only ancillas still in flight can change it). If a weight-11 undetected logical F contained a fault-free interval of at least $G = 52$ layers inside its time span (G follows from D and the longest check window; SM S2), no detector could see both the part of F before and the part after it, so both parts would be undetected with observables adding to a nonzero value, and one would be an undetected logical of weight ≤ 10, which is impossible. Hence F spans at most $10 \times 51 = 510$ layers. The noisy circuit is layer-periodic (inserting an 18-layer block at any layer after the first four gives the circuit with one more round), so fault-free blocks before and after F can be deleted without changing any detector that F touches, which maps F into a circuit with at most 32 rounds, excluded by the exact runs[C43].

Upper bound. A weight-12 data logical injected after the data reset fires no detector and flips the observable ($R = 1, 3, 6, 10$, both bases), so $d_{\text{circ}} = 12$[C42].

The lower bound rests on two earlier audited computations, $d_{\text{ext}} = 11$ for the Fig. 12 class[C11] and the completeness of the 4224-element list[C13]; a direct solve of the full DEM without the lift reduction did not terminate within 1–2 hours and is not part of the proof[C37].

How short can the period be? The circuit uses the same qubits as the IBM and SBB circuits (one ancilla per check) but 18 time steps per round instead of 8, i.e. 2.25 times as many steps and correspondingly more idle locations[C44]. Within colour-symmetric schedules with orders in the Fig. 12 class, 18 is optimal: for every period $P \leq 17$ (pipelined) and $P \leq 20$ (all checks sharing one reset/measure frame) every legal deterministic circuit has a weight-11 undetected error when the number of rounds is large enough. This follows from SAT encodings of the bulk frame model (two independent encodings for the pipelined case), whose lift assertions were confirmed on the exact DEM in all 991 sampled cases[C45]. So 18 (pipelined)

and 21 (non-wrapping) are the minimum periods of these families. This is a statement about many rounds, not about a fixed small R. A plausible explanation, not a proof, is that two of the six classes (X_0 and Z_2) need a CNOT window of at least 15 steps[C45]. The class of orders is not an arbitrary choice either: among all 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 or 9, with one order type per colour, Fig. 12's extended-code class is the only one with $d_{\text{ext}} \geq 11$ on both sides (all other assignments have verified extended logicals of weight ≤ 10)[C46]. With less symmetry the question is open: invariance under a 12-element translation subgroup excludes $P \leq 11$, and every subgroup of order ≥ 2 excludes $P = 8$, but per-check-free schedules with $P = 9, \ldots, 17$, and other subgroups with $P = 12, \ldots, 17$, are undecided[C47]. Earlier, the period-8 colour-symmetric circuits in the class were classified directly: all 944 legal deterministic ones have a weight-11 undetected error already at $R = 1$[C38].

Relation to prior work. To our knowledge no previous circuit for the gross code itself, with or without additional ancillas, has been shown, numerically or rigorously, to reach $d_{\text{circ}} = 12$[C40]. The morphing circuits of Refs. [3, 5], which use the same number of qubits and are believed to reach 12 [5], act on a different $[[144, 12, 12]]$ code and have no dedicated ancillas (below). Nor is ours the first BB circuit proven to reach the code distance: an exact value 6 for an IBM-type $[[72, 12, 6]]$ circuit is in the public data of Ref. [7], and we prove 6 for the IBM $[[72, 12, 6]]$ circuit (SM S8)[C21]. Nguyen, Rimbach-Russ and Bosco proposed lower-bounding interleaved circuits via a non-interleaved counterpart combined with a single-round reduction, as future work [19][C29]; our lift argument is a different route.

Why the original orders cannot reach 12. For Fig. 12 itself we prove SBB's conjecture: $d_{\text{circ}} = 11$ for any R, both bases. A complete symmetric depth-first search (DFS) on its extended code (360 merged columns per side, a 24-element order-preserving translation symmetry, 15 orbits) finds no logical of weight 10, reproduced on an independently constructed instance and under a second symmetry group, and weight 11 is attained[C10,11]. By Prop. 4 of SBB this is optimal among non-interleaved circuits[C12].

Retiming the same orders does not help in any family we can decide. For periodic schedules of period $P \leq 8$ the answer is exact: keeping Fig. 12's orders forces $P = 8$, a complete search finds 4 conflict-free circuits up to translation (Fig. 12 among them), and each has injection-verified weight-11 undetected errors for $R = 1, \ldots, 6$ in both bases[C14,15] (Fig. 2). For longer periods, a bulk frame-model analysis finds a weight-11 error of ten data faults and one hook in every legal deterministic circuit of several symmetric and per-check families (SM S7, Table S2)[C16,18]; other families remain open[C17]. The new circuit escapes this by changing the order within the pairs (and the timing) while keeping the pairs, and with them the extended code.

Circuit distance must be proven: morphing circuits. Morphing circuits [3] have no dedicated ancillas: each round contracts half of the stabilisers of a "mid-cycle" code onto single qubits, measures them, and re-expands, so that between rounds the state is encoded in an "end-cycle" code. Applied to the $[[288, 12, 18]]$ BB code with the homomorphism f_x, this yields a 288-qubit circuit whose end-cycle code is a $[[144, 12, 12]]$ code with weight-9 checks, reported with $\tilde{d}_{\text{circ}} \leq 12$ [3]; Ref. [5] (App. C) describes a morphing idling circuit "for the gross code" believed, from Monte-Carlo sampling, to reach 12. We prove $d_{\text{circ}} = 12$ for this circuit (public layer order) for every number of rounds, both bases, noisy or ideal boundaries[C1], by a single-round reduction (Prop. 1 of Ref. [4], whose premise we verify in a basis-free form, SM S5)[C3] and exhaustive search. This end-cycle code is not the gross code: its minimum stabiliser weight is 9, versus 6[C2]; whether it is the circuit of Ref. [5] is not stated there and we have not confirmed it. To our knowledge this is the first rigorous d_{circ} lower bound for any morphing circuit[C4].

The same construction has more freedom than its tables suggest: ten (mother code, homomorphism) pairs, two contraction variants and a free order of the terms (a_2, a_3), (b_2, b_3) give 80 circuits, which we reconstructed gate by gate and matched against an independent implementation. Their exact d_{circ}: 72 equal the end-cycle distance $\tilde{d}$, 8 fall below it (Fig. 3; Table S1)[C5]. For $[[108, 8, 10]] \to [[54, 8, 8]]$ the four legal term orders give 6, 6, 7, 7 (7 for the public implementation's order)[C6]: the caption of Table IX of Ref. [3] states that the order of the elements of A and B is not important for these schedules because the end-cycle codes are invariant under $a_2 \leftrightarrow a_3$, $b_2 \leftrightarrow b_3$; the codes are invariant, but the circuit distance is not[C6]. For f_x on the gross code the end-cycle code is $[[72, 12, 8]]$ but $d_{\text{circ}} = 6$, showing the bound ≤ 6 of that table to be tight[C7]. The sub-$\tilde{d}$ errors consist only of CNOT faults within a single interior half-cycle; they are not boundary artefacts[C8].

Discussion. The gross code admits a single-ancilla syndrome circuit that preserves its full distance: the obstacle identified by SBB is specific to non-interleaved circuits, not intrinsic to the single-ancilla model, although with the original CNOT orders we find it again in every retiming family we could decide. Two methodological points carry over. Keeping the extended code fixed while changing the order turns an open-ended circuit search into a finite lift problem against a precomputed list of minimum-weight extended logicals. And neither code-level invariances nor unrefuted upper bounds determine d_{circ}: the

morphing family contains circuits that fall short of their end-cycle distance in ways the code does not reveal; an exact criterion for morphing circuits will be reported separately. The same tools give further proven distances for BB circuits, among them $d_{\text{circ}} = 10$ for a depth-8 $[[108, 8, 10]]$ circuit (previously ≤ 8) and $d_{\text{circ}} = 6$ for the IBM $[[72, 12, 6]]$ circuit (SM S8, Table S3)[C19,21,22].

What a larger d_{circ} buys. d_{circ} is a guarantee of the circuit: no fewer than d_{circ} faults can cause an undetected logical error, whatever the decoder. How much of this reaches the logical error rate depends on p and on the decoder, and we have not evaluated it for the gross circuits. For two smaller pairs of circuits that differ in d_{circ}, leading-order structural estimates (not rigorous bounds) give a few-fold benefit at $p = 10^{-4}$: about 4–5 times for $[[54, 8, 8]]$ morphing circuits with $d_{\text{circ}} = 6$ versus 7, and about 10 times for the IBM $[[108, 8, 10]]$ circuit ($d_{\text{circ}} \leq 8$) versus our $d_{\text{circ}} = 10$ circuit; at $p \gtrsim 5 \times 10^{-4}$ this method does not determine the ratio, which grows more slowly than p^{-1}. With today's practical decoders (BP-OSD, Relay-BP) at p between 5×10^{-4} and 10^{-3}, the measured benefit for the $[[54, 8, 8]]$ pair is at most about 1.2–1.6 times (95% upper limits), because both decoders fail on a similar small fraction of weight-3 to weight-5 fault sets in the two circuits; for the $[[108, 8, 10]]$ pair it is undetermined[C48]. A larger d_{circ} is thus a prerequisite for the low-p scaling, not by itself a demonstrated performance gain.

Open questions. (i) What is the minimum period of a $d_{\text{circ}} = 12$ single-ancilla circuit for the gross code with less symmetry than ours, in particular per-check-free schedules with $P = 8, \ldots, 17$? (ii) Does the distance gain survive the extra idling of a longer period, i.e. is the logical error rate at accessible p and with practical decoders lower than for the depth-8 circuits? (iii) Do the proven distances survive correlated faults, leakage, or CZ-native compilations? (iv) Is the IBM morphing idling circuit of Ref. [5] the one we certify? (v) Three-round ($\mathbf{Z}_3$) morphing circuits and non-Abelian morphing codes [3] are not covered here.

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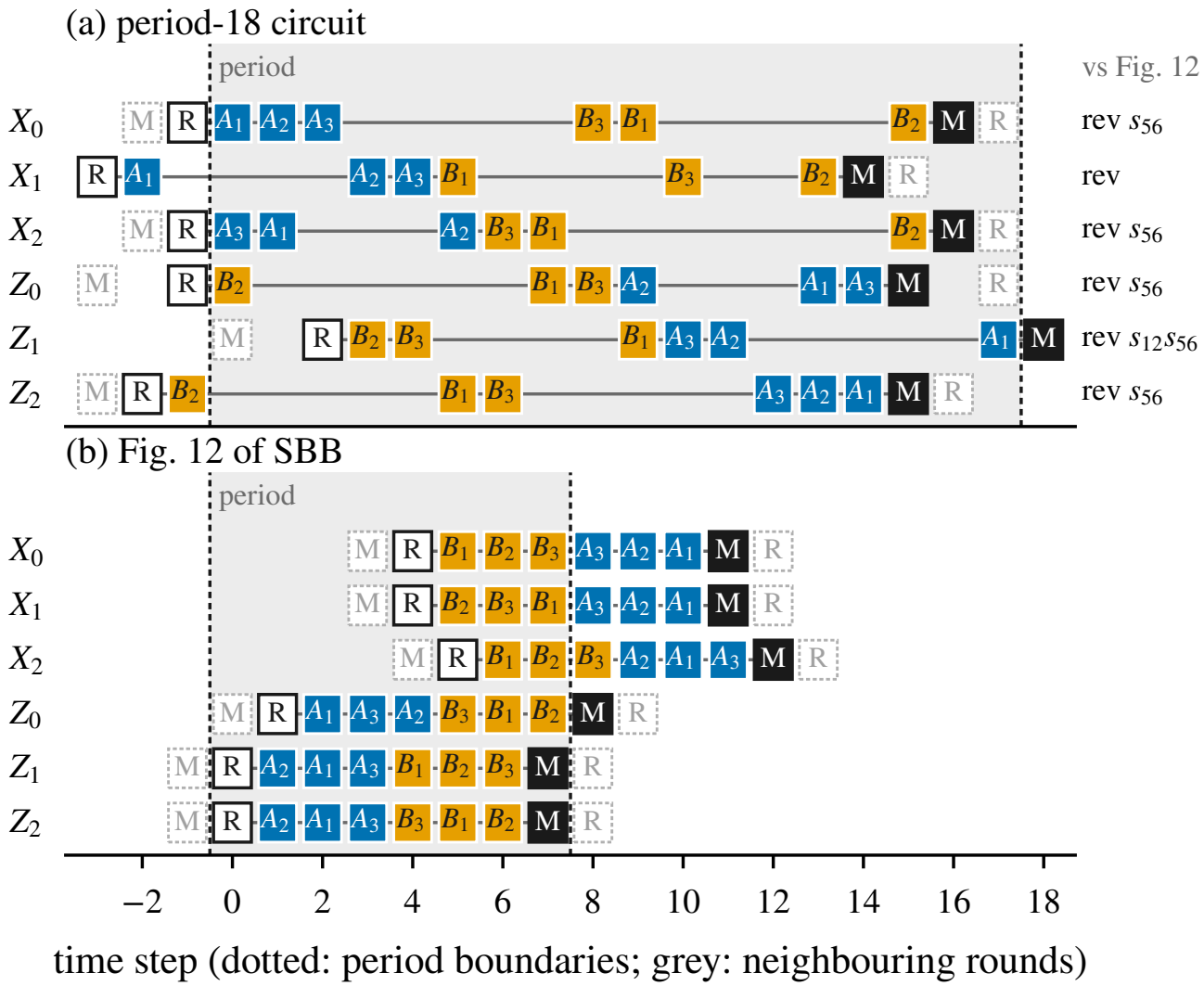


Figure 1: One period of (a) the period-18 circuit of Table 1 and (b) Fig. 12 of SBB [2], on the same time scale. Rows: the six colour classes ($(x + y)$ mod 3, 24 checks each); boxes: CNOTs, labelled by the term A_k (blue) or B_k (orange) they implement; R, M: ancilla reset and measurement. Windows that cross the period boundary are drawn continuing into the next period. (a) is pipelined: resets at steps $\{2, 15, 16, 17\}$, measurements at $\{0, 14, 15, 16\}$, and every check's window crosses a period boundary. Its class orders are obtained from those of Fig. 12 by reversal followed by s_{56} (X_0, X_2, Z_0, Z_2), by reversal alone (X_1), or by reversal, s_{12} and s_{56} (Z_1); all 144 checks change order, the extended code is identical. Per period, 96 data qubits switch between X- and Z-type CNOTs four times and 48 twice. (b) has period 8, is also pipelined, and is not interleaved (every data qubit switches twice). Circuit (a) has $d_{\text{circ}} = 12$ (SD6, both bases, every R); whether 12 is reachable at period 8 is open[C41,42,44].

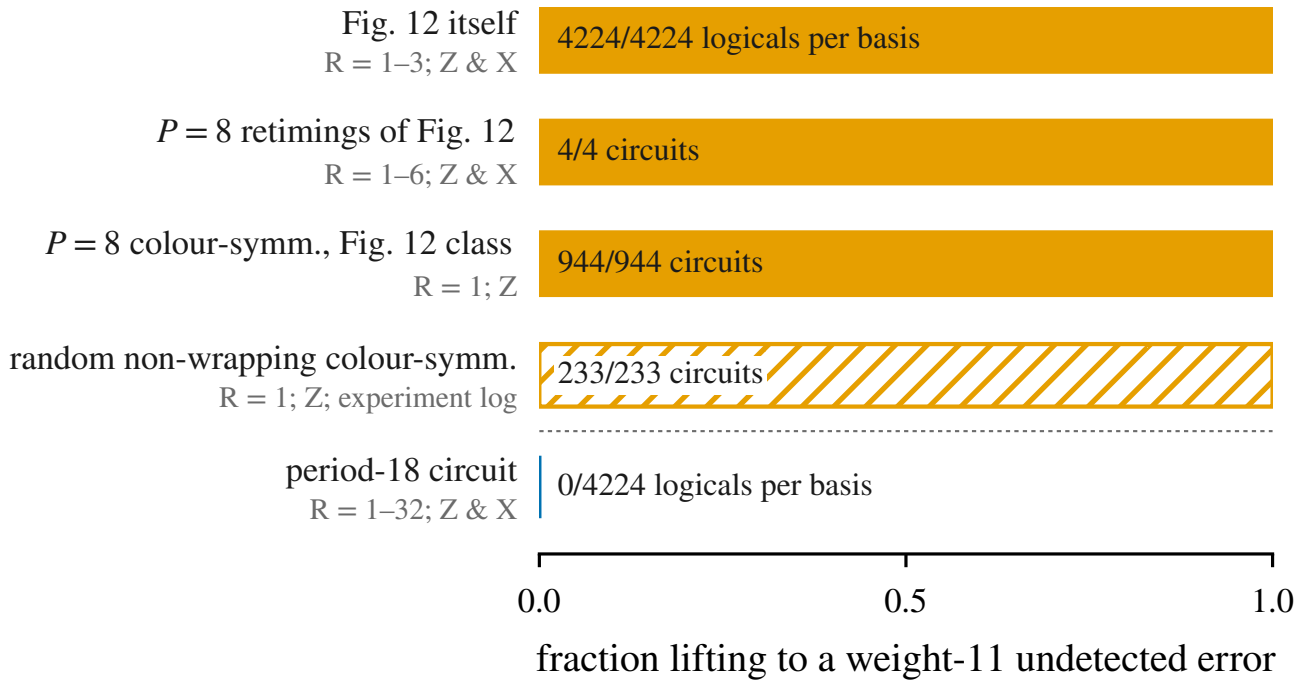


Figure 2: Exact lift outcomes. Bar: fraction of units with a weight-11 undetected circuit error. For Fig. 12 and for the period-18 circuit the units are the 4224 weight-11 extended logicals of one basis; for the other rows they are circuits (counted if at least one weight-11 undetected error exists). Fig. 12: 4224/4224 ($R = 1$–3, both bases)[C13,34]. The 4 conflict-free period-8 retimings of the Fig. 12 orders: 4/4, injection-verified, $R = 1, \ldots, 6$, both bases[C14,15]. Colour-symmetric period-8 circuits in the Fig. 12 class: 944/944 ($R = 1$, Z basis)[C38]. Random legal deterministic non-wrapping colour-symmetric circuits of the first search family: 233/233, none certified (hatched: control taken from the experiment log)[C39]. Period-18 circuit of Table 1: 0/4224 for $R = 1, \ldots, 32$, both bases (the experimenter's run to $R = 36$ agrees)[C42]. All counts are at finite R; the bulk obstructions for longer periods are in Table S2.

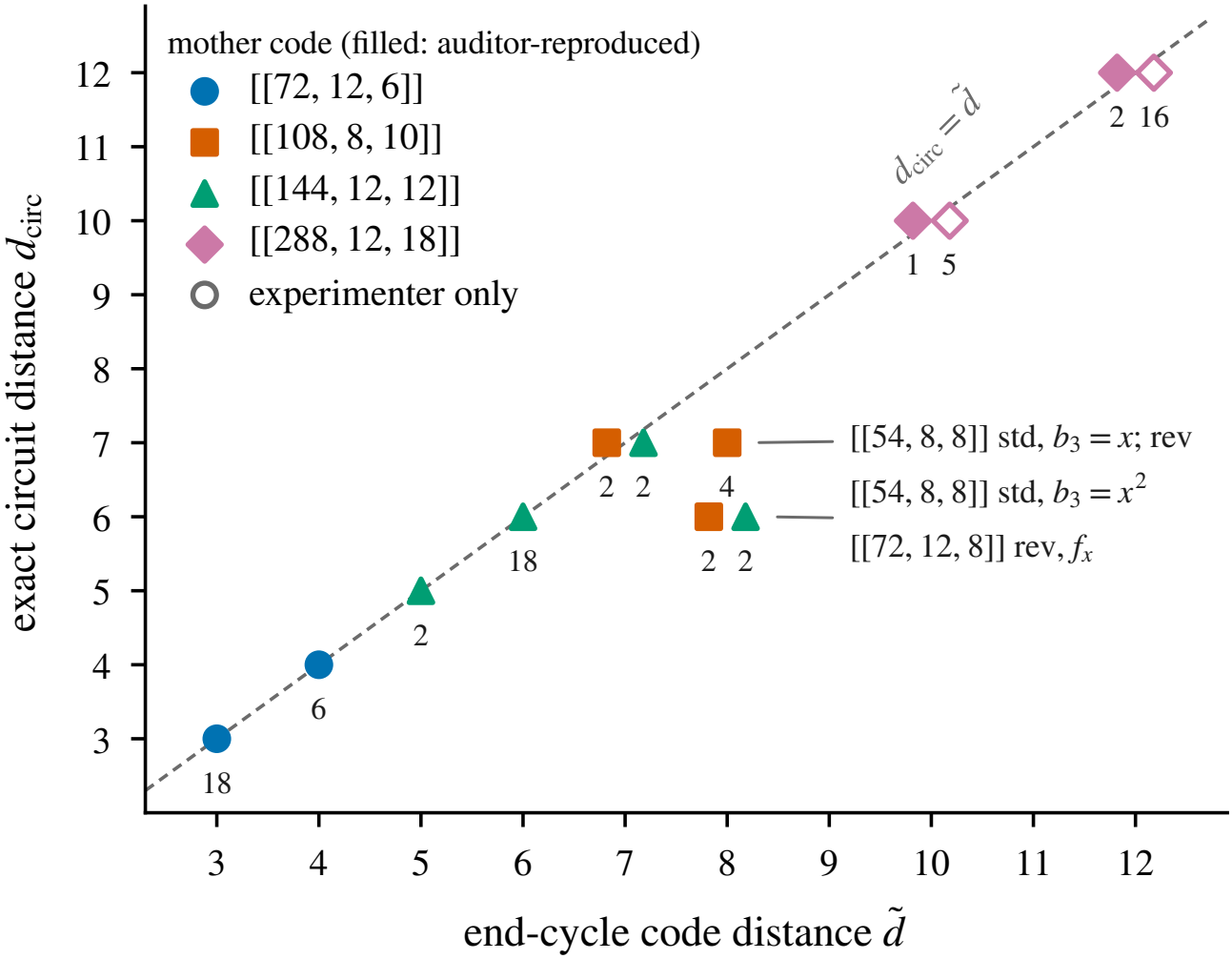


Figure 3: Exact circuit distance d_{circ} versus end-cycle distance $\tilde{d}$ for the 80 two-round morphing circuits of Ref. [3] (SD6; both memory bases give the same value). Numbers: circuits per marker. 72 circuits lie on the diagonal, 8 below it: for $[[108, 8, 10]] \to [[54, 8, 8]]$ the Table IV ("std") orders with $b_3 = x^2$ give 6, those with $b_3 = x$ and two Table VIII ("rev") variants give 7; for $[[144, 12, 12]]$, f_x, the two Table VIII variants with end-cycle code $[[72, 12, 8]]$ give 6, the bound ≤ 6 of Table IX of Ref. [3] being tight. The end-cycle codes are invariant under $a_2 \leftrightarrow a_3$, $b_2 \leftrightarrow b_3$; the circuit distance is not. Filled markers: independently reproduced (all 56 circuits with mother codes $[[72, 12, 6]]$, $[[108, 8, 10]]$, $[[144, 12, 12]]$; 3 of the 24 $[[288, 12, 18]]$ circuits); open markers: proven by the experimenter pipeline only[C5,6,7,9].

Supplemental Material

S1 Fault model and definition of d_{circ}

Noise is inserted by a fixed routine (SD6 [9]): after every CNOT a two-qubit depolarizing channel; after every reset a flip; before every measurement a flip; on every qubit not acted on in a time step a single-qubit depolarizing channel (also in reset and measurement layers). A *fault* is one Pauli component of one such channel. Two faults with identical detector and observable signatures are the same DEM mechanism; d_{circ} counts mechanisms. For lower bounds of CSS circuits we use the CSS relaxation: in a Z-basis memory we keep only Z-type detectors and the X part of each fault (and vice versa). This relaxation can only lower the minimum, so it is valid for lower bounds; upper bounds are always checked on the full circuit with all detectors. Lower bounds hold for any sub-model (e.g. no idle noise). For the morphing circuits the witnesses of $d_{\text{circ}} \leq \tilde{d}$ use idle components after a reset layer (or, in the first round, initial reset flips); for a no-idle model those upper bounds rely on the noisy initial data reset and would need re-checking otherwise[C28].

S2 The gross-code circuits with $d_{\text{circ}} = 12$

Circuits. Three single-ancilla circuits are certified, all colour-symmetric with orders in the Fig. 12 class: `cs_P18_H24` (period 18, pipelined; the circuit of Table 1; `search/experiments/e024a/cs_P18_H24.json`), `p18w` (period 18, pipelined, a different circuit; `search/experiments/e024b/p18w_spec.json`) and `cand_p21` (period 21, non-wrapping: all ancillas reset at step 0 and measured at step 20; `search/experiments/e021/cand_p21_spec.json`); two further non-wrapping instances of period 38 and 26 were certified first. The statements below are for `cs_P18_H24` unless noted; the others are summarised at the end.

Code and conventions. Gross code [1]: $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$, qubit and check index $i = 6a + b$ for the monomial $x^a y^b$. Our check matrices are element-wise equal to those of the benchmark implementation (not merely equivalent); $n = 144$, $k = 12$, $d = 12$ [1][C41]. Term labels: $A_1 = x^3$, $A_2 = y$, $A_3 = y^2$, $B_1 = y^3$, $B_2 = x$, $B_3 = x^2$. The X check at (x, y) acts on left qubit $(x, y) + A_k$ and right qubit $(x, y) + B_k$; the Z check at (x, y) on left qubit $(x, y) - B_k$ and right qubit $(x, y) - A_k$. Colour of a check: $(x + y)$ mod 3. Table 1 was extracted from the schedule file by `paper/scripts/extract_class_orders.py`, which asserts that this labelling reproduces every check's support, that each (type, colour) class has a single reset step, order, CNOT steps and measurement step, and that each class order is the image of the Fig. 12 order of the same class under an element of $\langle \text{rev}, s_{12}, s_{56} \rangle$ (right column of Table 1). The Fig. 12 orders in the same labelling are X_0: $B_1B_2B_3A_3A_2A_1$, X_1: $B_2B_3B_1A_3A_2A_1$, X_2: $B_1B_2B_3A_2A_1A_3$, Z_0: $A_1A_3A_2B_3B_1B_2$, Z_1: $A_2A_1A_3B_1B_2B_3$, Z_2: $A_2A_1A_3B_3B_1B_2$ (pipelined, period 8)[C42].

Legality. 72+72 checks, each with exactly six CNOTs on its support at distinct times inside its (reset, measure) window; no data qubit in two CNOTs at one time step; validated also by the benchmark's schedule checker; resets at steps $\{2, 15, 16, 17\}$ and measurements at $\{0, 14, 15, 16\}$ (windows wrap past the period boundary); the noiseless circuit fires no detector and the DEM builds without non-determinism; one dedicated ancilla per check, reset–six CNOTs–measure per round. Interleaving: over one period 96 data qubits switch four times and 48 twice (cyclically) between X- and Z-type CNOTs[C41].

Extended-code invariance. In a Z-basis memory only X errors matter. An X error on the ancilla of an X check after its k-th CNOT reaches the suffix $\{q_{k+1}, \ldots, q_6\}$; Z-check CNOTs (data→ancilla) do not change the data X frame. So each fault's final data X frame is zero, one data qubit, or a proper suffix of an X-check order, and its column is $(H_Z s, L_Z s)$. Modulo the check itself, a suffix of length 5 equals $\{q_1\}$, length 4 equals $\{q_1, q_2\}$, length 3 equals the complement of $\{q_1, q_2, q_3\}$, and length 2 is $\{q_5, q_6\}$. The column set is therefore determined by the three sets $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$, $\{q_5, q_6\}$ per check; reversal maps them to $\{q_5, q_6\}$, the last three, $\{q_1, q_2\}$, which give the same columns, and the swaps s_{12}, s_{56} leave them unchanged. The X basis is symmetric. Computed check: for every certified circuit both sides have 360 merged columns whose key sets equal those of Fig. 12[C32,42].

Lift reduction (template identity + Table I). We add one observable per row of H and of L on the final data readout. (i) In the circuit template the augmented observable of row H_i equals the sum of all detectors of check i (first round, consecutive-round comparisons, final readout), so for an undetected fault set the H parts of the images sum to zero and the logical effect equals the L part. (ii) Every nonzero image is one extended-code column (SBB Table I). Hence a weight-w undetected logical yields an extended logical of weight $\leq w$; with $d_{\text{ext}} = 11$, $w \geq 11$, and for $w = 11$ the eleven images are nonzero, distinct and form one of the listed weight-11 extended logicals. The original DEM's terms are unions of augmented terms, so the bound transfers. Prop. 3 of SBB is not used; neither interleaving nor pipelining affects (i) or (ii). Checked on the DEM for every R used: every term's per-check detector parity equals the H part of its image, the observable equals the L part, and the image set equals the 360 columns, both bases. In a Z-basis memory the template has no X-check detectors; fewer detectors can only enlarge the set of undetected fault sets, so the lower bound also holds with all detectors[C33].

Exact lift. Input: the 4224 weight-11 extended logicals per basis from the independent DFS of the E-018 audit, expanded by the 24-element group[C13]. For each logical, CP-SAT decides whether one DEM term per column can be chosen (exactly-one constraints) with every detector XOR zero. Positive control: Fig. 12 lifts 4224/4224. `cs_P18_H24`: 0/4224 for $R = 1, \ldots, 32$, both bases (64 runs); the experimenter's independent lift program agrees and extends to $R = 36$[C42].

All R: gap argument. Let $D = 18$. (1) Settling: if the last fault of a fault set $F_<$ is at layer a, the data frame produced by $F_<$ is constant from layer $a + D - 1$ on, since only ancillas still in flight can change it, and only through the remaining CNOTs of their current instance, all at layers $\leq a + D - 2$. (2) Splitting: let F be a weight-11 undetected logical with faults at layers $\leq a$ (part $F_<$) and $\geq b$ (part $F_>$) and none in between. A comparison detector of check c between instances $j - 1$ and j can respond to both parts only if the measurement of instance j is at a layer $\geq b$ and instance $j - 1$ starts at a layer $\leq a + D - 2$; with $\Delta_{\max} = 17$ the longest reset-to-measurement span, this is impossible when $b - a \geq 2D + \Delta_{\max} - 1 = 52$. The first-round and final readout detectors give the weaker condition $b - a \geq D + S + \Delta_{\max}$ with offset $S = 5$. If no detector sees both parts, each is undetected and their observables add to a nonzero value, so one is an undetected logical of weight ≤ 10, contradicting the reduction. Hence consecutive faults of F are less than $G = 52$ layers apart, and F spans at most $10 \times 51 = 510$ layers. (3) Deletion: the noisy SD6 circuit with $R + 1$ rounds equals the R-round circuit with an 18-layer block inserted at any layer $x \in [4, 18R]$ (checked layer by layer, both bases). A detector that straddles a deleted block equals the sum of the two original detectors; when the data frame is identical on both sides of the cut (the cut lies before the first fault, $x + D \leq f_1$, or after the frame has settled, $x \geq f_{11} + 2D - 2$), deleting fault-free blocks changes no detector touched by F. This maps F into a circuit with at most 32 rounds, which the exact lift excludes. The argument needs no premise on the schedule beyond those checked; it was checked line by line by the auditor[C43].

Upper bound. Weight-12 data logicals found by CP-SAT were injected as flips after the data reset: for $R = 1, 3, 6, 10$, both bases, zero detectors fire and the observable flips; removing any one qubit makes the error detectable[C42].

The other certified circuits. `p18w`: legal, deterministic, pipelined (resets $\{0, 1, 2, 17\}$, measurements $\{0, 1, 16, 17\}$), interleaved; 24 checks keep their Fig. 12 order, 120 use other class members; extended code equal to Fig. 12; lift 0/4224 for $R = 1, \ldots, 24$, both bases; weight-12 injection. Its arbitrary-R step uses a "slab" version of the splitting argument (one fault-free slab of layers between the earliest reset and the latest measurement of a round), which for pipelined circuits needs a premise: on every data qubit, every propagating CNOT of round k inside the overlap with round $k + 1$ precedes every reading CNOT of round $k + 1$. The premise was machine-checked for $R = 3, 8, 24$, all boundaries, both bases (0 violations), after which $R \leq 23$ suffices[C43]. `cand_p21`: non-wrapping, so rounds are cleanly separated and the splitting argument needs one fault-free round; lift 0/4224 for $R = 1, \ldots, 14, 16$ and a solver-free meet-in-the-middle check for $R \leq 3$; $R \leq 13$ suffices[C31,34,35]. Heuristic minimum-weight searches on `cand_p21` and the period-38 instance found nothing below 12[C36].

Not done. A direct minimum-weight solve of the full DEM without the lift reduction (MaxSAT 7200 s, CP-SAT 3600 s, $R = 1$, Z, on the period-38 instance) did not terminate. The $d_{\text{ext}} = 11$ computation was reproduced under two symmetry reductions but not without symmetry[C11,37].

Minimum period of the colour-symmetric families. Family: one unit per (type, colour $(a + b)$ mod 3); per unit one of the eight class orders, a start $s_u \in [0, P)$ (non-wrapping: $s_u = 0$), distinct CNOT times in $[s_u + 1, s_u + P - 2]$ and the measurement at $s_u + P - 1$; legality (six distinct times mod P per data qubit) and determinism are encoded exactly. Lazy clauses forbid every frame-model-liftable modelled weight-11 extended logical; 168 of 352 representatives with overlapping realisations are not modelled, which can only weaken the clauses and is safe for UNSAT. Results: pipelined $P = 8, \ldots, 17$ UNSAT (a second, independent encoding with absolute times agrees for $P = 8, \ldots, 17$); non-wrapping $P = 20$ UNSAT; final CNFs re-solved from scratch by independent SAT solvers (three for $P = 20$ non-wrapping and $P = 12$; two for $P = 17$). Completeness of the family encoding: at $P = 8$ it enumerates exactly the 944 circuits found by independent backtracking (hash-identical), and 75/75 independently sampled legal deterministic circuits at $P = 12$, 17 (pipelined) and 20 (non-wrapping) are models. Soundness of the clauses against the exact DEM: 991/991 sampled lift assertions (including multi-hook and paired columns) and 99/99 clause assertions confirmed. UNSAT means that every legal deterministic circuit of the family has a weight-11 undetected logical when the number of rounds is large enough; it is not a finite-R statement. Restricting the CNOT window of class X_0 or Z_2 to ≤ 14 steps is UNSAT at $P = 24$, whereas the other four classes can each be so restricted; since a window is at most $P - 3$, this suggests $P \geq 18$, but the minimum is established by the direct runs, not by this observation[C45].

Uniqueness of the order class. For a check, the extended-code columns are the data columns plus three residual sets (P_1, T, P_3); up to reversal there are 90 types per side, of which 45 contain a "bad triple" (a residual of length 3 which, with nine data columns, forms a weight-10 extended logical; verified for all 10 bad triples on both sides). For each of the 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8, 9 (3, 4, 3, 12, 1, 1 subgroups), one type per colour, a SAT search with witness clauses (every witness re-verified from scratch as ≤ 10 distinct columns of that assignment's extended code with vanishing H part and nonzero L part) leaves only translates of Fig. 12: 48/48 instances (24 colourings,

two sides) are UNSAT once those are blocked, and Fig. 12 appears exactly in $(a+b)$ mod 3 and its refinements. Not covered: per-check-free assignments and colourings of index 12, 18, 24, 36[C46].

Less symmetry (partial). Pipelined circuits invariant under the order-12 subgroup $\langle(1,2)\rangle$ have the bulk obstruction for every $P \leq 11$; for every subgroup of order ≥ 2, $P = 8$ is excluded. These were spot-checked by re-running ($P = 10$ for $\langle(1,2)\rangle$; $P = 8$ for $\langle(6,3)\rangle$ and $\langle(0,3)\rangle$), not fully re-audited. Per-check-free schedules and the remaining subgroups at $P = 12, \ldots, 17$ are undecided[C47].

Period-8 colour-symmetric circuits in the Fig. 12 class. At $P = 8$ the offsets are forced to 1..6 and measurement to 7; each class chooses one of the eight orders and a start in $\mathbf{Z}_8$. Two independent enumerations give the same 944 legal deterministic circuits; an exact lift at $R = 1$, Z basis, finds a weight-11 undetected logical in each (944/944), each witness rechecked[C38].

Search and classical control. The circuits were found by SAT models over colour-symmetric schedules whose clauses block every modelled weight-11 extended logical in the bulk frame model; with the Fig. 12 orders pinned the model is UNSAT, consistent with SM S7. Control in the non-wrapping family of the first search: the same incremental SAT sampler without the blocking clauses produced 2277 legal deterministic circuits, none blocking all modelled logicals, and exact lifting of a random 10% (233) certified none[C39]. Further counts from the experiment logs (how many reasoned candidates certified, timings) are not audited and are not used.

S3 Extended code, Prop. 3, and the exhaustive search

Extended code. For a single-ancilla circuit in which check c applies its CNOTs in the order $i_1, \ldots, i_w$, the residuals are the suffixes $\{i_l, \ldots, i_w\}$, $l = 2, \ldots, w$. The Z-type extended code appends each Z-check residual (as the sum of its data columns) to (H_X, L_X); analogously for X. Zero columns are dropped and identical (H, L) columns merged. Prop. 3 of Ref. [2]: $d_{\text{circ}} \geq d_{\text{ext}}$ for any single-ancilla CNOT circuit (interleaved or not); the converse $d_{\text{circ}} \leq d_{\text{ext}}$ (their Cor. 1) needs non-interleaving and is not used for our lower bounds. The key step—each single fault, propagated to the end of the circuit, has a data image equal to zero or to exactly one extended-code column—was checked on the evaluated circuits: $[[108, 8, 10]]$ ($R = 3$, 16 416 fault components per basis), gross Fig. 12 ($R = 1, 2, 3$), IBM $[[72, 12, 6]]$ ($R = 1, \ldots, 10$) and the four LRC theorems: 0 exceptions, all columns realised[C27].

DFS. To decide whether $H_e x = 0$, $L_e x \neq 0$ has a solution with $|x| \leq W$: a minimum-weight solution is syndrome-minimal; translate it so that it contains the representative of its lowest orbit; branch on the columns of an odd syndrome row, forbidding earlier siblings (a partition of the search space); prune by depth $+ \lceil |s|/w_{\max} \rceil > W$; look up the last column in a hash table. The symmetry group is computed (translations that are code automorphisms, map each check's CNOT order to the image check's order, and preserve logical classes, verified by solving $L \circ T = ML + NH$ with rank $M = k$). Validation: exact solution counts against unpruned meet-in-the-middle enumeration on 22 BB72 instances and on > 64-row direct-sum instances; split-run completeness (sum of split runs = unsplit run); runs without symmetry for $[[108, 8, 10]]$ (both sides, $\approx 2.5 \times 10^{10}$ nodes, 0 solutions) and for the four LRC theorems[C19,22].

$[[108, 8, 10]]$ *circuit.* Terms indexed $[A_1, A_2, A_3, B_1, B_2, B_3]$ with $A = (x^3, y, y^2)$, $B = (y^3, x, x^2)$, $(\ell, m) = (9, 6)$. X checks: reset at 0, CNOTs at 1–6, measure at 7; term order $[0, 1, 2, 5, 3, 4]$ on even y rows and $[2, 1, 0, 5, 3, 4]$ on odd rows. Z checks: reset at 4, CNOTs at 5,6,7,0,1,2, measure at 3; term order $[3, 4, 5, 2, 1, 0]$ on even rows and $[3, 4, 5, 0, 1, 2]$ on odd rows. (The Z side applies the B block first; the reversal acts on the A block.) No data-qubit conflicts, non-interleaved and partitionable. $d^X_{\text{ext}} = d^Z_{\text{ext}} = 10$ exactly (270 columns per side, 10 orbits of 27; weight-9 DFS: 0 solutions with and without symmetry; weight 10 attained); injection-verified weight-10 witness[C19].

Gross Fig. 12. 360 merged columns per side (144 of H_e-weight 3, 144 of weight 6, 72 of weight 9); 24 translations ($a + b \equiv 0$ mod 3) preserve every check's CNOT order and the logical classes; 15 orbits. Weight-10 DFS: 0 solutions on the original instance ($\approx 2 \times 10^{10}$ nodes per side), on an independently built lattice-coordinate instance, and under an order-8 subgroup (45 orbits, $\approx 5 \times 10^{10}$ nodes per side). Weight-11 solutions realised as 11 circuit faults for $R = 1, 2, 3$, both bases[C10,11]. A run without any symmetry ($\approx 7 \times 10^{11}$ nodes) was not performed.

S4 Exact lift test and arbitrary R for the IBM $[[72, 12, 6]]$ circuit

Let F be an undetected logical with $|F| = 5 = d_{\text{ext}}$. By the Table-I property each fault image is zero or one column; cancelling pairs leaves an extended logical T with $|T| \leq |F|$. Since no extended logical of weight ≤ 4 exists, all five images are nonzero and distinct and form a weight-5 extended logical. We list all of them (36 for the X-type, 252 for the Z-type extended code; meet-in-the-middle and DFS agree), group the DEM terms by the image of every circuit location (checked to list all locations: location counts equal those from `stim`), and search for one term per column with zero detector sum and nonzero observable. Result: 0 for $R = 1, \ldots, 10$, both bases; positive control at weight 6 finds solutions[C21]. *Arbitrary R.* (1) The CNOT windows of consecutive cycles are separated. (2) If an undetected logical F has an empty round strictly inside its time span, it splits into two undetected parts whose observables add, one of them a lighter undetected logical; for

$|F| = 5 = d_{\text{ext}}$ this contradicts Prop. 3, so a weight-5 logical spans at most 5 consecutive rounds. (3) The noisy $(R+1)$-round circuit equals the R-round circuit with one block inserted, so such a logical translates into a circuit with ≤ 7 rounds, covered by the exact runs. Hypotheses (1) and (3) and the detector structure were machine-checked; step (3) is a written argument[C21].

S5 Morphing circuits: construction, single-round instance, fold

Construction. Mid-cycle code $[[288, 12, 18]]$, $\ell = m = 12$, $A = x^3 + y^7 + y^2$, $B = y^3 + x + x^2$; homomorphism f_x; $K = \ker f_x$, $|K| = 72$. Each round: reset the 144 previously measured qubits, apply the 3 layers of the inverse contraction and the 3 layers of the next contraction (each layer a perfect matching of 144 CNOTs), measure 144 qubits. Every contracting stabiliser lands on a single same-basis measured qubit; the end-cycle codes have 72+72 weight-9 generators, rank 66+66, $k = 12$, qubit degree 5. Our construction is gate-identical to an independent one for all 3 homomorphisms $\times$ 2 layer orders $\times$ 2 bases $\times$ 2 starting rounds. Detectors are derived as the GF(2) kernel of noiseless samples and split into time-local blocks D_t[C1].

Instance. The set of DEM terms equals the set of projected fault signatures ($T = 1..5$, both bases). Every round-r fault touches only D_r, D_{r+1}. For $T = 1..5$ every round's instance is equivalent to the round-1 instance (time shift plus a translation for even r; rank conditions and a bijection on projected faults). The 72 translations by K are automorphisms of every layer and induce column permutations preserving undetected logicals; 12 orbits of 72[C1].

Fold condition (basis-free form). Let μ map each round-r fault to its time-reversed partner in round $r-1$ (checked to be a fault for every round-r fault). Let Z_r be the span of detector functionals that vanish on all faults of rounds $< r$. If F is undetected with last round r and round-r part E, then E is annihilated by Z_r, and $F' = (F \setminus E) \oplus \mu(E)$ is undetected with the same logical value whenever

$$\Psi(E) = \big[H_{\text{all}}(E) + H_{\text{all}}(\mu E);\ L(E) + L(\mu E)\big] = 0 \quad \text{for all } E \text{ annihilated by } Z_r, \tag{S1}$$

i.e. whenever every row of Ψ lies in the row space of Z_r restricted to round-r faults. A local version replaces Z_r by the functionals in span(D_r, D_{r+1}) vanishing on round-$(r-1)$ faults (a subspace, hence stronger) and depends only on two rounds. Both versions hold for $r = 2..T$, $T = 2..5$, both bases, including the noisy final round. Iterating the fold gives, for every T, an undetected logical of no larger weight inside one round, i.e. a solution of the single-round instance. The arbitrary-T step therefore rests on exact checks for $T \leq 5$ plus periodicity of the local condition and of the single-round instance (machine-checked hypotheses plus a written argument). *Remark.* The per-fault statements in the proof of Prop. 1 of Ref. [4] ("same detectors", "same observables") do not hold literally in a general detector basis, and a naive block-wise test fails at the final round; the condition above is the one the fold needs and it holds[C3].

Search and witness. Weight-11 DFS, all 12 orbit cases, both bases: 0 solutions on two equivalent instances (sparse basis: $\approx 2.8 \times 10^{10}$ nodes per basis; record basis, different tree: $\approx 2.0 \times 10^{11}$ nodes per basis). The end-cycle codes have distance exactly 12 (DFS) and minimum stabiliser weight exactly 9; injecting a weight-12 end-cycle logical as 12 single-qubit faults after a reset layer gives 0 detectors and flipped observables for every admissible round and T[C1,2].

S6 The 80-circuit morphing family

Table S1: Exact d_{circ} of the 80 two-round morphing circuits of Ref. [3] (SD6, both bases equal). $\tilde{d}$: end-cycle distance. Variants: Table IV ("std") or Table VIII ("rev"), with the four $(a_2, a_3)/(b_2, b_3)$ orders.

mother	end-cycle (variants)	#	d_{circ}	$= \tilde{d}$?
$[[72, 12, 6]]$	$[[36, 12, 3]]$	18	3	yes
	$[[36, 12, 4]]$ (rev)	6	4	yes
$[[108, 8, 10]]$	$[[54, 8, 8]]$ std, $b_3 = x^2$	2	6	**no**
	$[[54, 8, 8]]$ std, $b_3 = x$	2	7	**no**
	$[[54, 8, 8]]$ rev	2	7	**no**
	$[[54, 8, 7]]$ rev	2	7	yes
$[[144, 12, 12]]$	$[[72, 12, 6]]$	18	6	yes
	$[[72, 12, 5]]$ (rev)	2	5	yes
	$[[72, 12, 7]]$ (rev, f_y)	2	7	yes
	$[[72, 12, 8]]$ (rev, f_x)	2	6	**no**
$[[288, 12, 18]]$	$[[144, 12, 12]]$	18	12	yes
	$[[144, 12, 10]]$ (rev)	6	10	yes

Inventory. 10 (mother code, homomorphism) pairs: $[[72, 12, 6]]$ (f_x, f_y, f_{xy}), $[[108, 8, 10]]$ $(f_y$ only), $[[144, 12, 12]]$ (f_x, f_y, f_{xy}), $[[288, 12, 18]]$ (f_x, f_y, f_{xy}); $\times$ {Table IV, Table VIII} $\times$ 2 orders of (a_2, a_3) $\times$ 2 orders of (b_2, b_3). Reconstructed from Tables IV, VIII and IX of Ref. [3] only, and found gate-identical (80/80) to an independent implementation; end-cycle distances agree; Table IX of Ref. [3] fixes the permutations of the Table VIII circuits it lists[C5].

Verification depth. The 56 circuits with mother codes $[[72]]$, $[[108]]$, $[[144]]$ were solved directly on the full $T = 4$ circuit (noisy boundaries) in both bases by two independent pipelines (different detector derivations, DEM-based versus own fault enumeration, different solvers, plus a tree-free meet-in-the-middle check on five instances): 112/112 agree. For the 8 sub-$\tilde{d}$ circuits the values were recomputed in 96 settings (SD6 versus no idle noise; full $T = 4$; $T = 5$ with noise only in half-cycles 2–4; $T = 5$ with noise only in one interior half-cycle): identical values throughout. Minimum witnesses consisting only of CNOT faults in one interior half-cycle were injected into circuits carrying all X- and Z-type detectors (16/16 pass)[C5,8]. The 24 circuits with mother code $[[288, 12, 18]]$ were proven by one pipeline (single-round DFS plus fold check, all flags verified); three of them (f_x rev with $\tilde{d} = 10$; f_y std and f_{xy} rev with $\tilde{d} = 12$) were reproduced independently on single interior half-cycle instances in both bases, and one also on the full $T = 3$ circuit without folding. The arbitrary-T statement for these circuits relies on the per-circuit fold computation (code and output flags audited, not independently re-implemented)[C9].

Details moved from the main text. The 6/6/7/7 split is by b_3 in the first round ($b_3 = x^2$: 6; $b_3 = x$: 7). Table IX of Ref. [3] itself lists only $(\ell, m) = (6, 6)$ and $(12, 6)$, so $[[54, 8, 8]]$ is not an entry of it; the statement concerned is its caption about Table IV schedules. The $[[72, 12, 7]]$ circuit (f_y) is tight at 7, and all 24 circuits derived from $[[288, 12, 18]]$ preserve their end-cycle distance (12 or 10)[C6,7,9].

Wording constraints we observe. We do not claim that "the circuit of Table II of Ref. [3] has $d_{\text{circ}} = 6$", because Tables II/IV do not fix the order of (b_2, b_3). We do not attribute the sub-$\tilde{d}$ errors to reset faults: CNOT-only realisations exist[C6,8]. The $[[72, 12, 8]]$ circuits are listed in Table IX of Ref. [3] with bound ≤ 6; our result is tightness[C7].

S7 Gross code with the Fig. 12 orders: lift structure, period-8 family, bulk obstructions

This section concerns circuits that keep the CNOT orders of Fig. 12. The frame model described here is also the one used to design the circuits of SM S2. Fig. 12 itself: $d_{\text{circ}} = 11$ (main text). Bulk summary (moved from the main text): the frame model agrees with the exact lift in the bulk but not at the initial time boundary[C18]; every legal deterministic circuit keeping the Fig. 12 orders has a bulk weight-11 error of ten data faults and one hook for translation-symmetric non-wrapping circuits of any period, symmetric pipelined circuits with $P = 8, \ldots, 12$ ($P = 12$ by a single encoding), several subgroup-invariant pipelined families at $P = 9, 10$, and per-check-free non-wrapping circuits with $P \leq 16$ (Table S2)[C16]. In these families the certificate has one form: with the Fig. 12 orders, the timing conditions that would hide every such hook error from the detectors are mutually inconsistent.

Weight-11 logicals. Exhaustive enumeration with two different DFS programs gives 4224 weight-11 extended logicals per side, all in free orbits of the 24-element group (176 orbits); 360 of them consist of ten data columns and one length-2 residual ("10 data + r2"). An exact lift checker augments the circuit with one observable per row of H and L on the final data readout, so each DEM term carries its extended-code image; for each extended logical a frontier dynamic program over the 11 column groups decides liftability exactly. On Fig. 12 all $4224/4224$ lift for $R = 1, 2, 3$ (independently reproduced; the original run covers $R = 1..6$)[C13].

The benchmark family ($P \leq 8$). Rules: one period $P \in [2, 8]$ shared by X and Z checks; per check one reset, six CNOTs at distinct offsets $0 < o_1 < \cdots < o_6 < o_{\text{meas}} < P$ and one measurement; each data qubit in at most one CNOT per time step mod P; determinism checked on the circuit. Weight 6 forces $P = 8$, $o_k = k$, $o_{\text{meas}} = 7$; keeping the Fig. 12 orders leaves only $s_c \in \mathbf{Z}_8$ per check. A non-SAT backtracking search (forward checking, pigeonhole per data qubit, no symmetry) finds exactly 4 solutions with one start fixed and $32 = 8 \times 4$ without; two independent SAT encodings agree. All four are colour-uniform, deterministic, non-interleaved; one is Fig. 12. The other lattice isomorphism of Fig. 12 is related by a code relabelling, so the statement does not depend on that choice. Weight-11 witnesses were injected through the benchmark's own verification routine: 72/72 (4 circuits $\times$ $R = 1..6$ $\times$ 2 bases) fire no detector and flip an observable, with 11 distinct DEM terms[C14,15]. Not covered: other CNOT orders; $R \geq 7$ for this finite-R statement.

Frame model. A data X error at time θ is first seen by Z check c (readout times $T_c + nP$) in instance $H_c = \min\{n : T_c + nP > \theta\}$. Three readouts of a data qubit give three length-P open intervals; pairwise intersection is a difference constraint on the H_c, and by one-dimensional Helly a common time exists, at which an SD6 fault location is available. A hook (ancilla X after the fourth CNOT of an X check) fixes its arrival. Feasibility of the difference constraints (no negative cycle) implies a genuine weight-11 undetected error in the bulk; this was checked by 75 stim injections. The converse is not used. The model agrees with the exact lift checker in the bulk but not at the initial time boundary: on a $P = 16$ circuit, 96 logicals blocked in the model lift at $R = 2, 4, 6$ via round-0 detectors[C18].

Certificates. Blocking every "10 data + r2" logical requires a negative cycle for each; CP-SAT (path encoding, and

independently an unrolled Bellman–Ford encoding) and a CEGAR SAT loop (lazy determinism clauses; nogoods from the potentials of liftable logicals, with 24 translated images) give INFEASIBLE/UNSAT for the families in Table S2[C16].

Table S2: Bulk obstruction status for retimings of Fig. 12 that keep its per-check CNOT orders. "Proven" means: every legal deterministic circuit of the family has a bulk weight-11 undetected error, hence $d_{\text{circ}} \leq 11$ for sufficiently many rounds.

family	status	evide
symmetric (24-translation) non-wrapping frame, any P	proven	CP-S
symmetric pipelined, $P = 8, 9, 10$	proven	two C
symmetric pipelined, $P = 11$	proven	CP-S
symmetric pipelined, $P = 12$	proven, single encoding	CP-S
H-invariant pipelined, $P = 9$: $\lvert H\rvert = 12$ (3), 8, 6 (3), 4 ($\langle(0,3),(6,0)\rangle$, $\langle(3,3)\rangle$), 3	proven	CEG
H-invariant pipelined, $P = 10$: $\lvert H\rvert = 12$ $\langle(1,2)\rangle$, 8, 6 $\langle(2,4)\rangle$	proven	CEG
per-check-free non-wrapping frame, $P \leq 16$	proven, one encoding	CP-S
$\lvert H\rvert = 4$ $\langle(3,0)\rangle$ pipelined, $P = 9$	open	UNK
symmetric pipelined, $P \geq 13$	open	–
all other pipelined families with $P \geq 9$ (incl. $\lvert H\rvert = 2$, per-check-free, unlisted subgroups at $P = 10$)	open	per-ch
per-check-free non-wrapping frame, $P > 16$	open	–
different CNOT orders	open	–

[C16,17] *Classical control.* Over 2000 SAT-enumerated and $\approx 1.1 \times 10^5$ random-walk legal deterministic per-check $P = 9$ circuits, all 720 single-hook logicals remained liftable in the frame model. This is a classical baseline, not a proof[C17].

S8 Other BB codes

Table S3: Proven circuit distances (all results of this work). "Prior" is the best value previously reported for the same circuit or code. All entries: SD6 fault set, both memory bases, any number of rounds R.

code / circuit	prior	d_{circ}	tag
gross $[[144, 12, 12]]$, period-18 interleaved (Table 1)	≤ 10 (IBM), 11 (Fig. 12)	**12**	[C41]
gross $[[144, 12, 12]]$, second period-18 and period-21 circuits	–	12	[C31,43]
gross $[[144, 12, 12]]$, Fig. 12 of [2]	11 (conj.)	**11**	[C10]
$[[288, 12, 18]] \to [[144, 12, 12]]$ morphing (f_x; not the gross code)	≤ 12 [3]	**12**	[C1]
80 two-round morphing circuits	$\leq$ bounds [3]	$72 = \tilde{d}$, $8 < \tilde{d}$	[C5]
$[[108, 8, 10]]$, y-parity two-colour	≤ 8 [1]	**10**	[C19]
$[[72, 12, 6]]$, IBM circuit [1]	6 (conj.)[a]	6	[C21]
$[[72, 12, 6]]$, two-colour LRC	–	6	[C22]
coprime $[[126, 12, 10]]$, uniform LRC	≤ 9 [10]	9	[C22]
$[[120, 8, 12]]$ $(\ell, m) = (20, 3)$, uniform LRC	–	10	[C22]
$[[124, 10, 10]]$ $(31, 2)$, uniform LRC	–	9	[C22]

[a] An exact value 6 for one basis, $R = 6$, and a CZ/SI1000 compilation is contained in the public data of Ref. [7]; see text.

Results (moved from the main text). For $[[108, 8, 10]]$ the IBM circuit has $d_{\text{circ}} \leq 8$ [1, 3], and while the single-check bound for partitionable non-interleaved circuits does not exclude 10 for this code, uniform depth-8 tilings are limited to ≤ 9[C24]. A two-colour left-right circuit that reverses the CNOT order within one term block on checks of one y-parity (the X side on odd rows, the Z side on even rows; depth 8, one ancilla per check, all checks every round) has $d_{\text{circ}} = 10 = d$[C19]. To our knowledge this is the first $d_{\text{circ}} = 10$ circuit for this code and the first circuit distance for a BB code with $d \geq 10$ that comes with a stated, reproducible proof (here BB means $\ell, m \geq 2$; Ref. [17] lists $d_{\text{circ}} = 9$, method unstated, for a $[[54, 2, 10]]$ generalized-bicycle code); rigorous circuit distances exist for other qLDPC families [11, 12, 13], and the idea of position-dependent CNOT orders is established [14, 15, 2][C20].

For the IBM $[[72, 12, 6]]$ circuit [1], which is interleaved, $d_{\text{ext}} = 5 < 6$, so Prop. 3 alone does not suffice. The exact lift test shows that none of the weight-5 extended logicals lifts, for $R = 1, \ldots, 10$, and a splitting argument extends this to every R: $d_{\text{circ}} = 6$ in both bases, settling IBM's conjecture for this circuit and fault model[C21]. Public data of Ref. [7] already contain an exact value 6 for one basis and $R = 6$ of a CZ/SI1000 compilation of an IBM-type circuit; our additions are arbitrary R, both bases, the CNOT circuit, and the method[C21].

The same pipeline—colour the checks by a translation subgroup, prune block orders by single-colour-class witnesses, enumerate legal states, and prove the survivors by DFS—gives the theorems in the lower half of Table S3 [C22] and rigorous negative results: every two-colour LRC of the gross code has $d_{\text{circ}} \leq 10$, and every two- or three-class mod-3 LRC of $[[90, 8, 10]]$ has $d_{\text{circ}} \leq 8$[C23]. Apart from $[[72, 12, 6]]$, these are, to our knowledge, the first circuit distances for BB codes with $d \geq 10$ that come with a stated, reproducible proof; Ref. [17] reports $d_{\text{circ}} = 9$ for $[[70, 6, 9]]$ and $[[102, 22, 9]]$ BB/GB codes and for a $[[54, 2, 10]]$ GB code without describing how these values were obtained[C25].

Codes and orders. $[[72, 12, 6]]$ and $[[108, 8, 10]]$ as in Ref. [1] ($A = x^3+y+y^2$, $B = y^3+x+x^2$); coprime $[[126, 12, 10]]$ ($\ell = 7$, $m = 9$, $A = 1 + \pi + \pi^{58}$, $B = 1 + \pi^{13} + \pi^{41}$, $\pi = xy$; Ref. [10], Table 2); $[[120, 8, 12]]$ with $(\ell, m) = (20, 3)$, $A = 1 + x^3y^2 + x^7y^2$, $B = 1 + x^{16}y + x^9y$ (found in our own code screen; its equivalence to the $[[120, 8, 12]]$ code of other works is [UNVERIFIED]); $[[124, 10, 10]]$ with $(\ell, m) = (31, 2)$, $A = 1 + x^{27}y + x^{17}$, $B = 1 + x^{30}y + x^{12}y$ (own screen). Term orders (indices $[A_1, A_2, A_3, B_1, B_2, B_3]$): $[[72, 12, 6]]$ two-colour: as for $[[108, 8, 10]]$; cp126 uniform: X $[1, 0, 2, 4, 3, 5]$, Z $[4, 3, 5, 1, 0, 2]$; $[[120]]$ and $[[124]]$ uniform: X $[0, 1, 2, 3, 4, 5]$, Z $[3, 4, 5, 0, 1, 2]$. Each theorem: DFS at $W = w - 1$ with and without symmetry (0 solutions), DFS at $W = w$ finds solutions, injection-verified weight-w witness for $R = 1, 2, 3$ and by the time-translation argument of SM S4 for every R[C22]. All 864 proper two-colour LRC states of $[[72, 12, 6]]$ have $d_{\text{ext}} = 6$[C22].

Negative results. Gross: for all three index-2 colourings, every two-colour LRC has a verified weight-10 extended logical, hence $d_{\text{circ}} \leq 10$ (Cor. 1; the template is non-interleaved). $[[90, 8, 10]]$ ($A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$, $(15, 3)$): every LRC whose block order is constant on the $h = 0$ class and on its complement, or on each of the three cosets, for all four $\mathbf{Z}_3$ homomorphisms, has $d_{\text{ext}} \leq 8$[C23]. For partitionable non-interleaved circuits (circuits that can be cut in time into alternating X-only and Z-only CNOT segments), $[[90, 8, 10]]$ has $d_{\text{circ}} \leq 9$ and $[[108, 8, 10]]$ has $d_{\text{circ}} \leq 10$ by the single-check argument of Prop. 4 of Ref. [2]; uniform tilings give ≤ 8 ($[[90]]$, any depth) and ≤ 9 ($[[108]]$, depth 8). IBM-type pipelined circuits are pairwise non-interleaved but not partitionable and are not covered[C24].

S9 Prior work and novelty caveats

- Gross $d_{\text{circ}} = 12$ (search date 2026-10-10; arXiv API on titles/abstracts, keyword searches, the qldpc-challenge issues): no circuit for the gross code itself (single- or multi-ancilla, flag or Shor-style) with $d_{\text{circ}} = 12$ shown numerically or rigorously was found. The open question of Ref. [2] (v2, Sec. IV.3) is quoted verbatim in the text; it carries no depth condition, and the paper does not conjecture impossibility. Its App. H.2 envisages "non-staggered, interleaved variants" of the depth-8 circuit, i.e. the depth-8 question we leave open. Best previous values for the gross code itself: ≤ 10 for the IBM circuit and its 936 depth-7 variants [1], ≤ 10 in Ref. [6], 11 for Fig. 12. The morphing "12" of Refs. [3, 5] concerns a different code; Ref. [19] lists lower bounds for interleaved circuits as future work. Evidence strength medium-high; there is no external reproduction yet[C40]. The period reduction from 21 to 18, the minimum periods of the colour-symmetric families and the uniqueness of the order class among coset colourings are follow-up results of this work (2026-10-11) and were not part of that literature search[C45,46].

- Gross Fig. 12 $= 11$: Ref. [2] v2 still states 11 as a conjecture and suggests exhaustive methods without executing them; no proof of any gross-circuit $d_{\text{circ}} \geq 11$ was found in the literature checked (search date 2026-10-07)[C12].

- IBM $[[72, 12, 6]]$: partial precedent. The public results table of Ref. [7] contains an exact value 6 (two exact solvers) for a Z-basis, $R = 6$, CZ+H/SI1000 circuit of IBM type; it is not stated as a result in the paper and its gate-by-gate identity with the IBM order was not checked. The GitHub leaderboard [20] lists an upper bound 6/6 for $R = 6$[C21].

- $[[108, 8, 10]]$: literature values are ≤ 8; position- or colour-dependent CNOT orders have precedents [14, 15, 2]; the circuit is an instance of the left-right framework of Ref. [2]. Rigorous circuit-distance results exist for hypergraph-product codes [11], layer codes [13], and as certificates for amplified codes [12]; an algebraic LP framework gives upper bounds [16][C20].

- Exact circuit distances for BB/GB codes with $d \geq 9$ by others: Ref. [17] lists $d_{\text{circ}} = 9$ (without "$\leq$") for $[[70, 6, 9]]$ BB, $[[102, 22, 9]]$ GB and $[[54, 2, 10]]$ GB codes; the method is not stated anywhere in the paper (v1, the only version; its Tables 9–10 and Sec. IX.2 define d_{circ} and list the schedules, and "$\leq$" is used only for values quoted from other works). We therefore restrict our wording to proofs that are stated and reproducible, and to $d \geq 10$ with $\ell, m \geq 2$. Ref. [7] reports that exact solvers completed only the smallest BB circuit of its benchmark[C20,25].

- Morphing: no rigorous lower bound on d_{circ} for any morphing circuit was found (search date 2026-10-08); Ref. [18] certifies code distances of related constructions, not circuit distances[C4]. A dedicated search (2026-10-09) found no prior exact circuit distance for any morphing circuit, no prior evaluation of the 80-circuit family beyond the BP-OSD upper bounds of Ref. [3], and no prior result on retimings of the Fig. 12 circuit (whether interleaved circuits can reach

12 is stated as open in Ref. [2]). Ref. [19] proposes combining the non-interleaved lower bound of Ref. [2] with the single-round reduction of Ref. [4] as future work, without numerical results[C4,29,30].

S10 Reproduction

A single script re-runs the DFS lower bounds for $[[108, 8, 10]]$, the four LRC theorems, IBM $[[72, 12, 6]]$ and (optionally) gross Fig. 12; a full run passed on 2026-10-07; options add the morphing results (E-017, E-019) and the period-8 retiming family of Fig. 12 (E-020), passed on 2026-10-10. The script's `--with-gross12` option re-runs the period-21 gross $d_{\text{circ}} = 12$ proof (passed from a clean directory on 2026-10-11), and the `--with-gross12-p18` option re-runs the period-18 proof of Table I (rebuilt weight-11 list, exact lift for $R = 1, \ldots, 32$ in both bases, weight-12 injection and the gap-argument premises; passed from a clean directory on 2026-10-12, using only the auditor's scripts). For the period-18 circuits, the minimum-period runs and the uniqueness replay, the audit lists the commands (`research/audits/2026-10-11-E-024b.md`); the period-21 and period-8 commands are in `research/audits/2026-10-10-E-021.md`. Each audit report lists its own reproduction commands; witnesses for negative results are archived. [TODO: public repository and DOI – PI decision.]